Study Guide: Distributed System Architectures and Edge Computing
This study guide provides a comprehensive review of distributed systems, covering centralized and decentralized architectures, peer-to-peer (P2P) networking models, and the emerging field of edge computing.

Part 1: Short-Answer Quiz
Instructions: Answer the following questions in 2-3 sentences based on the provided materials.

1. **What are the three components of the traditional three-layered logical view of an application?**
2. **How does the basic client-server model operate regarding process roles and communication?**
3. **What is the fundamental difference between a two-tiered and a three-tiered architecture?**
4. **In a structured P2P system, how are nodes organized and how is information located?**
5. **What is the "partial view" concept in unstructured P2P systems?**
6. **Compare the communication efficiency and speed of Random Walk versus Flooding search methods.**
7. **What specific functions might be assigned to a "Superpeer" in a hybrid P2P network?**
8. **How does a BitTorrent client use a "tracker" to facilitate file sharing?**
9. **What is the primary essence of edge-server architecture?**
10. **Why is edge orchestration considered trickier than managing resources in the cloud?**

Part 2: Answer Key

1. The traditional view consists of the **user-interface layer** (units for the UI), the **processing layer** (application functions without specific data), and the **data layer** (the actual data manipulated by the client). This layering is common in distributed information systems using traditional database technology.
2. The model consists of **servers** (processes offering services) and **clients** (processes using services), which can reside on different machines. Clients follow a **request/reply model**, sending a request to the server and waiting for a reply once the service is provided.
3. A **two-tiered architecture** typically involves a client and a single server configuration, often splitting layers between the two machines. A **three-tiered architecture** places each of the three logical layers (UI, processing, and data) on a separate, dedicated machine.
4. Structured P2P systems organize nodes in a specific **overlay network**, such as a logical ring or a hypercube, where nodes are responsible for services based on their ID. These systems provide a **LOOKUP(key)** operation to efficiently route requests to the correct associated node.
5. In unstructured systems, each node maintains a **partial view** consisting of a small subset of other nodes in the network. Nodes periodically select a peer from this view to

exchange membership information, which helps maintain both randomness and robustness in the overlay.

6. **Random walks** are more communication-efficient because they query neighbors sequentially or in limited parallel paths, but they take longer to find results. **Flooding** reaches more nodes quickly by broadcasting to all neighbors, but it is less efficient and consumes significantly more network resources.
7. **Superpeers** are selected nodes appointed to perform special functions, such as maintaining an index for search operations or monitoring the state of the network. They can also act as intermediaries to help other peers set up connections.
8. A tracker is a server that keeps an accurate, real-time account of active nodes (the **swarm**) that possess specific chunks of a file. When a user opens a torrent file, the tracker provides the references needed to connect with those nodes and begin trading data.
9. Edge-server architecture involves deploying systems where servers are placed at the **edge of the network**, specifically at the boundary between enterprise networks and the Internet. This physical placement is intended to improve performance for real-time applications by decreasing the distance data must travel.
10. Edge orchestration is complex because it requires managing **resource allocation** to guarantee availability across diverse locations and **service placement** for mobile applications. Additionally, the system must perform **edge selection**, as the geographically closest infrastructure is not always the optimal choice for offering a service.

Part 3: Essay Questions

Instructions: Use the source context to develop detailed responses for the following prompts. (Answers not provided).

1. **The Evolution of Centralized Architectures:** Discuss the transition from single-tiered "dumb terminal" configurations to modern three-tiered architectures. How does the physical deployment of the user-interface, processing, and data layers impact system scalability?
2. **Search Determinism in P2P Systems:** Compare the deterministic nature of "LOOKUP" operations in structured P2P systems (like Chord or Tapestry) with the probabilistic "searching" required in unstructured networks. Under what conditions would a developer choose one over the other?
3. **The Mechanics of Unstructured Overlays:** Analyze the mathematical trade-offs between "Search Size" (S) and replication (r) in unstructured P2P networks. How do variations like limited or probabilistic flooding attempt to mitigate the overhead of standard broadcasting?
4. **The Rationales for Edge Computing:** Critique the three main drivers for edge computing: latency/bandwidth, reliability, and security/privacy. Based on the source text, which of these drivers are supported by reality, and which are based on potentially false assumptions?

5. **Hybridization and Superpeers:** Explain how hybrid P2P systems utilize a two-layer topology management approach. How does the interaction between the lower layer (random partial views) and the upper layer (selective references) create a more efficient network than a purely unstructured model?

Part 4: Glossary of Key Terms

Term	Definition
BitTorrent	A collaboration principle where users search for files via global directories, obtain torrent files, and trade data chunks with a swarm of peers.
Chord / Koorde	Examples of structured P2P algorithms used to organize nodes and manage data lookups.

Data Layer	The logical layer containing the actual data that a client wants to manipulate through application components.
DHT (Distributed Hash Table)	A structured P2P model (e.g., Mainline DHT) used by systems like BitTorrent for content-addressable networking.
Edge Computing	An architecture where servers are placed at the boundary between internal enterprise networks and the Internet to reduce latency.

Flooding	A search method where a node sends a lookup query to all its neighbors, who then forward it to all of their neighbors.
Hypercube	A specific logical structure used to organize nodes in certain structured P2P systems.
Overlay Network	A virtual network where connections are set up between nodes at the application level, rather than the physical level.

Partial View	A list of other nodes maintained by a peer in an unstructured network to facilitate random exchanges and network robustness.
Random Walk	A search method where a node randomly selects a neighbor to query; if that neighbor doesn't have the answer, it randomly selects one of its own neighbors.

Request/Reply Model	The standard communication pattern in client-server architectures where a user process waits for a service provider's response.
Superpeer	A node in a hybrid P2P system appointed with special functions, such as indexing, monitoring, or connection setup.
Swarm	The collection of active nodes that possess chunks of a specific file and are available for trading in BitTorrent.

Tracker	A central server in the BitTorrent ecosystem that maintains an account of which nodes are active in a swarm.